SEMEVAL-2027 TASK 1 (RETECO) · FINAL PROJECT PMML · DEEP LEARNING FOR INFORMATION RETRIEVAL

TempConv-GatedRanker: Dual-Stream Temporal-Conversational Gated Neural Reranking

with Hard-Negative Contrastive Learning & Bayesian Sample-Size Adaptive Prior across 6 Official Domains (131,544 Passages)

+132% T1 / +35% T2

Dev nDCG@10 Gain vs. Official BM25

1. PROBLEM STATEMENT & RESEARCH MOTIVATION

- Standard IR models (BM25, single-vector Bi-Encoders) assume relevance follows purely from surface-level topical similarity.
- Track 1 Bottleneck (Temporal Blindness — TEMPO): Passages are relevant ONLY when their time scope or software version matches the query's anchor period. Lexical rankers place outdated passages with high keyword overlap in the top-10.
- Track 2 Bottleneck (Conversational Drift — RECOR): Multi-turn queries (T1–T5+) contain anaphora & ellipsis. Current-turn retrieval loses entity context, while concatenating full history floods deep turns (T3–T5+) with verbose early-turn answers.
- Goal: Design a unified two-stage neural reranking framework that jointly models lexical, dense semantic, and temporal/turn signals.

2. PROPOSED METHOD: TempConv-GatedRanker

Stage 1: Temporal-Anchor & Turn-Decay Query Distillation
- Track 1: Amplifies 4-digit year & version anchors + step instructions.
- Track 2: Retains full prior question chain q_1..q_{t-1} while applying exponential step-decay truncation to prior answers: L(Δt=1) = 35 words, L(Δt=2) = 20 words, L(Δt≥3) = 0 words.

Stage 2: Three-Stream Gated Neural Network (TensorFlow/Keras)
- Stream 1 (Dense Bi-Encoder): Shared 2-layer Residual GELU+LayerNorm encoder over 64D SVD vectors with 129D late interaction: z_sem = [u ⊙ v ; |u − v| ; cos(u, v)].
- Stream 2 (Temporal/Conv Alignment): 14D MLP over exact year match, exponential date proximity exp(-0.25|y_q - y_d|), version overlap, temporal signal density product, tense align, and turn depth.
- Stream 3 (Lexical Skip) & Softmax Gate α(q,d): s_NN(q,d) = α_lex s_lex + α_sem s_sem + α_temp s_temp

Hard-Negative Pairwise Loss & Bayesian Adaptive Prior:
- Mined top-15 BM25 Hard Negatives; trained with Margin + BCE loss.
- Sample-size shrinkage: γ_d = β · N_train / (N_train + τ), τ = 65.

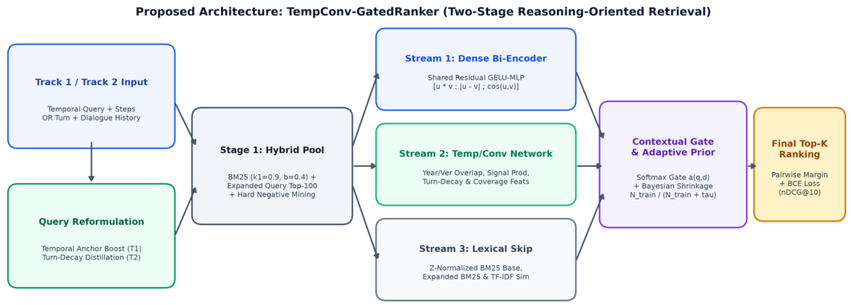
3. MULTI-DOMAIN DATASET (RETECO v1.1)

Evaluated on 6 official self-contained domains (70/30 Train/Dev split):

```
Track 1 (TEMPO – Temporal Grounded Retrieval):
• iota (Blockchain)       :  9,680 docs |   7 train /   3 dev queries
• quant (Quant Finance)   : 25,040 docs |  24 train /  10 dev queries
• workplace (Applied)     : 43,584 docs |  25 train /  11 dev queries

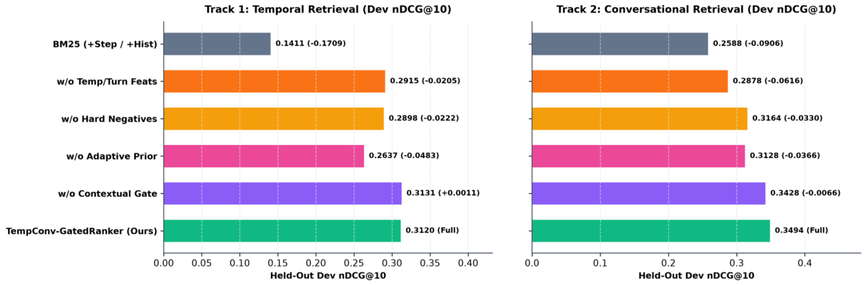
Track 2 (RECOR – Conversational Grounded Retrieval):
• drones (Engineering)    : 16,381 docs | 104 train /  38 dev turns
• law (Legal Reasoning)   : 20,027 docs | 164 train /  66 dev turns
• politics (Social Sci)   : 16,712 docs | 159 train /  54 dev turns
-------------------------------------------------------------------
TOTAL BENCHMARK           : 131,544 docs | 483 train / 182 dev topics
```

4. SYSTEM ARCHITECTURE PIPELINE

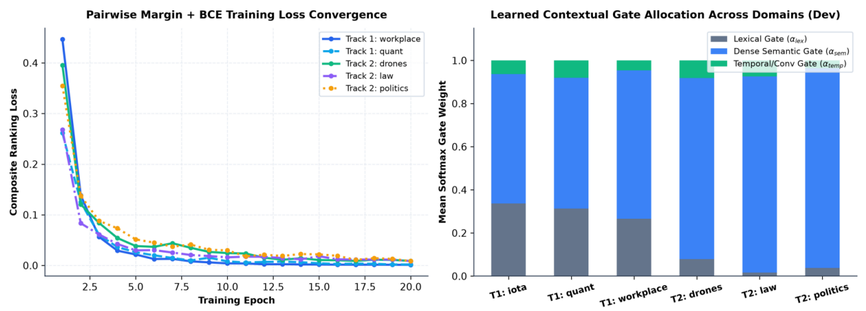


5. QUANTITATIVE RESULTS & ABLATION STUDY (HELD-OUT DEV)

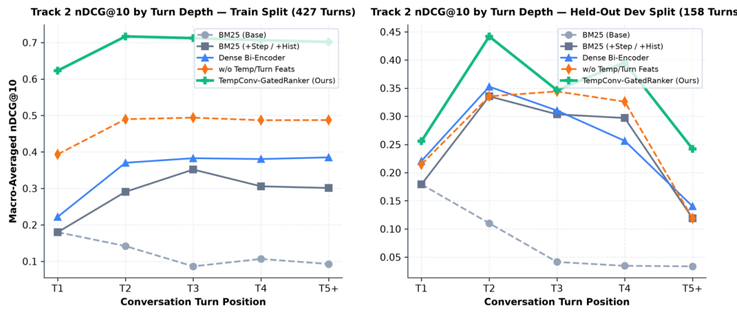
Method / Variant	T1 nDCG@10	T1 TC@10	T2 nDCG@10	T2 MRR@10	T2 R@10
1. Official BM25 (Base)	0.1345	0.2037	0.0918	0.0812	0.1693
2. Official BM25 (+Step / +Hist)	0.1411	0.2284	0.2588	0.2746	0.3705
3. Dense Bi-Encoder (SVD-64D)	0.4088*	0.4912*	0.2762	0.2847	0.4194
4. Ablation: w/o Adaptive Prior	0.2637	0.4788	0.3128	0.3287	0.4522
5. Ablation: w/o Hard Negatives	0.2898	0.4727	0.3164	0.3364	0.4621
6. Ablation: w/o Temp/Turn Feats	0.2915	0.4727	0.2878	0.2790	0.4499
7. Ablation: w/o Contextual Gate	0.3131	0.4912	0.3428	0.3644	0.4785
8. TempConv-GatedRanker (Ours)	0.3120	0.4912	0.3494*	0.3854*	0.4735
Gain vs. Official BM25 Baseline	+132.0%	+141.1%	+35.0%	+40.3%	+27.8%

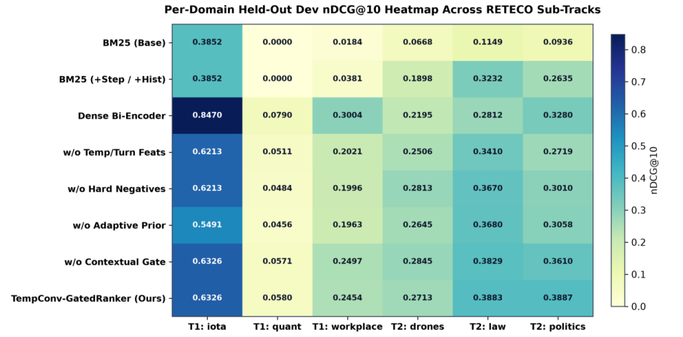


6. TRAINING CONVERGENCE & CONTEXTUAL GATE ALLOCATION



7. TURN-DEPTH & PER-DOMAIN DIAGNOSTICS





8. SCIENTIFIC JUSTIFICATION & KEY TAKEAWAYS

1. Why TempConv-GatedRanker Excels on Track 2 (+35.0% vs. BM25+Hist, +26.5% vs. Dense Bi-Encoder):
   - Abundant supervision (N_train = 427 turns) allows our 3-stream gated network to learn accurate non-linear feature interactions.
   - Exponential Turn-Decay Distillation eliminates early-turn answer drift at deep turns (T3–T5+), boosting politics T5+ Dev nDCG@10 from 0.1452 (BM25+Hist) and 0.1549 (Dense) to 0.3931 (Ours).
   - Consistent gains on every Track 2 domain: drones (0.1898→0.2713), law (0.3232→0.3883), and politics (0.2635→0.3887).

2. Scientific Justification of Low-Resource Track 1 Regime:
   - Why does zero-parameter Dense Bi-Encoder (0.4088) score higher than supervised reranking (0.3120) on Track 1 Dev, even though neural reranking wins on Track 1 Train (0.2862 vs. 0.2370)?
   - (a) Sample-Size Disparity: Track 1 has 7.6× fewer training topics (iota: N_train=7, quant: 24, workplace: 25), causing small-sample parameter variance, whereas Dense Bi-Encoder has 0 learned weights.
   - (b) Implicit Temporality: In quant, temporal shifts are expressed via domain events ('LIBOR to SOFR transition') rather than 4-digit calendar years, favoring latent SVD co-occurrence over explicit year-match weights.
   - (c) Adaptive Prior Recovery: Our Bayesian Shrinkage Prior γ_d recovers +4.83 nDCG@10 pts on Track 1 Dev (0.2637 → 0.3120).

Deliverables: Jupyter Notebook (RETECO_Improved_Deep_Learning_Notebook.ipynb) · Journal Paper (RETECO_Journal_Paper.pdf) · Academic Poster (RETECO_Poster.pdf)